

ARCHIT RITESH JAIN

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Education

Vishwakarma Institute of Technology

August 2019 – May 2023

Bachelor of Technology in Production Engineering [GPA – 9.27]

Pune, Maharashtra

Relevant coursework includes Mechatronics and Controls, Robotics, Product Development, Engineering Design, Mechanical Design, Manufacturing Processes, Data Science, Computer Programming

Experience

Kalyani Strategic Management Services Technologies Solution Pvt Ltd

June 2024 – Present

Robotics Engineer

Pune, Maharashtra

- Collaborating cross-functionally to deploy robotic solutions in a factory environment, driving end-to-end development of mission-level autonomous systems. My work spans the perception-to-control pipeline, with a focus on real-time autonomy, fleet interoperability, and architecture design for a robust communication system.
- Designed a full-stack autonomy solution for robotic platforms such as quadrupeds (unitree go2, b2), humanoids (untiree g1), and wheeled mobile robots. Using a modular autonomy stack with 3D LiDAR-Inertial Odometry (LIO) mapping & localization in ROS 2, along with multi-terrain navigation for indoor, semi-structured, & factory environments.
- Built & deployed Hardware Abstraction Layers (HAL) to ensure portability across robot configurations & standardize control interfaces. Integrated these robots into a web-based fleet software developed over the MERN stack.

e-Yantra, ERTS Lab, Indian Institute of Technology Bombay

June 2023 – June 2024

Senior Project Technical Assistant

Mumbai, Maharashtra

- Researched grasping algorithms for picking unknown objects in utilization with a two-finger gripper UR5 robotic arm. Developed an Elliptical-based Centroid Grasp method using point cloud segmentation. Formulated a unified ROS-based pipeline integrating Dexnet, Height Assistive Feature, and Grasp Net algorithms to facilitate seamless development.
- Worked on integrating a mobile robot and a UR5 robotic arm, working collaboratively to pick and efficiently sort packages within a warehouse for packaging. Implemented using ROS with SLAM, Nav2 Stack and MoveIt tools.
- Delivered hands-on embedded systems training to 100+ faculties nationwide and instructed 1300 students in robotics.

Flytbase Labs Pvt. Ltd

January 2023 – May 2023

Robotics Engineer Intern

Pune, Maharashtra

- Led the development of an autonomous drone operations platform for Beyond Visual Line of Sight (BVLOS) missions. Designed middleware components running on edge devices embedded in drones for real-time communication & control.
- Developed micro-services for autonomous waypoint navigation, failsafe execution, and health monitoring, communicating over MQTT and integrated with FlytBase's cloud platform.
- Implemented mission planning modules using MAVROS and PX4 APIs, enabling real-time trajectory updates, mission re-routing, recovery mechanisms including battery-critical RTL, GPS loss detection, and failsafe state transitions.

Projects

Robot Actuator Joint Design | Gear Design, Torque Calculations, Material Selection

July 2022 - October 2022

- Designed and fabricated a compact, high-torque actuator for robotic arms operating under heavy-load conditions.
- The actuator integrates a NEMA-17 stepper motor coupled with a 10:1 custom gear reduction system, achieving a peak torque output of 42 kg·cm at the elbow joint.

Humanoid Robot | Mechanical Design, Stress Analysis, Machining, 3D Printing

May 2021 - October 2021

- Led the mechanical design of a humanoid robot's upper body with a focus on compact form factor. Drafted mechanical drawings with required GD & T standards. Developed a 2-DOF parallel manipulator for neck movement, with sub-degree repeatability. Designed custom linear actuators for the shoulder joint using a 10:1 gear reduction system.

Skills

Robotics:: ROS 1, ROS 2, SLAM (LIO/Visual), Mavros, Nav2 Stack, MoveIt, Gazebo, RViz, OpenCV, PCL, URDF, SolidWorks, 3D Printing, Edge Computing (Jetson, Intel NUC, Arduino, RaspberryPi,)

Languages: Python, C, C++, Javascript, Shell Script

Utilities: LATEX, Markdown, Git, Docker, GitHub Actions, Linux

Achievements / Extracurricular

- Achieved 1st place in the Smart India Hackathon - Hardware Edition 2022, recognized by the Indian Ministry.
- Secured AIR 5 in the DD National ABU Robocon 2022, securing the Best Software Award.
- Certified SolidWorks Additive Manufacturing Associate by Dassault Systèmes.
- Conducted trainings and workshops on various robotics domains.
- Volunteered in social welfare activity to record audiobooks for blind school students.